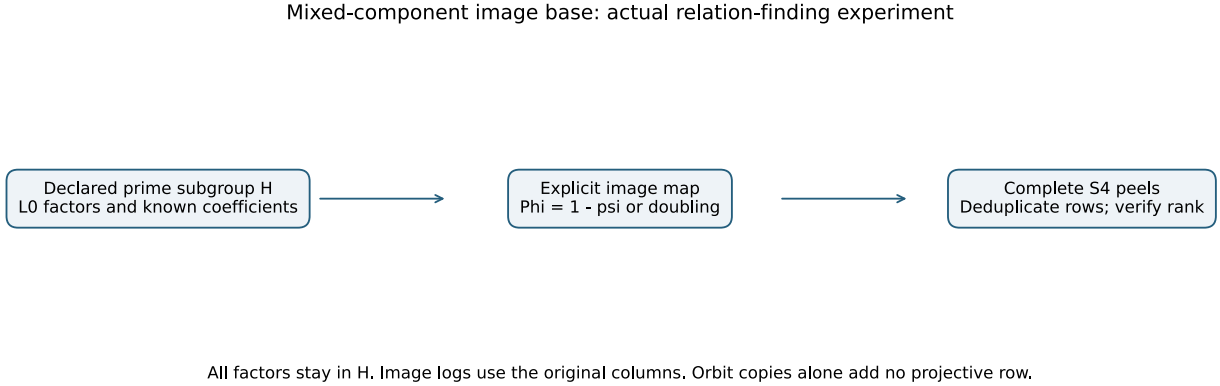
Validation status of the twist image-base control study



Scientific results and repository software gates are separate obligations. The following records preserve failures and interruptions as well as passes.

Scientific verification

Requirement	Recorded outcome
Frozen input integrity	Exact SHA-256 checked before execution
Original 189 cases and deterministic A/A	All exits 0, byte-identical non-timing trees
Doubling control	All 21 cases, reference and polynomial first/A/A complete
Independent signed sets, matrices and recovery	210 passing audits; 189 full-rank target/factor recoveries and 21 deliberately inactive-column censored controls
Frozen baseline parity	All 621 queries of seven old S4 prefixes equal in sets, rank and system count
Negative controls	Omitted relation, changed input hash, corrupt eigenvalue, changed coefficient and false Lean theorem rejected
Image coordinate identity	Exact integer remainder zero, IDC1h7e0104841476f369; pinned Lean 4.34.1 conditional theorem and positive axiom audit
Cold resources and phase sums	105 complete same-origin eleven-execution timed rounds plus 630 non-timing records; all 210 cost rows admitted, all deterministic traces and exact sums checked
Resume control	Complete units preserved, incomplete unit rejected, partial inputs retained, copied bytes unchanged
External complete GB	378 fixed systems × eight passes = 3,024 equal bases; raw input and S-pair reductions check every pair

These are known-answer fixtures with scalar 271. Source auditing excludes use of that scalar by the candidate; physical isolation of the scalar input is not established. The supplied public coordinate encodings are used for alignment while every required lift is recomputed and charged.

Software gates and retained failures

The final serial native suite passes 57/57 with one Rayon and one test thread. An earlier parallel run passes 56/57: unchanged shared F4 test hilbert_stop_gives_the_same_reduced_basis_as_the_plain_run reports cheaper=19, below threshold 20. The failure is retained without weakening the test. This study does not edit src/cryptanalysis/f4_fp.rs.

The local parallel library run passes 4,482 tests, fails six in unchanged jv_cover, and ignores 119. Its serial continuation was interrupted by migration; no pass is credited. First managed gate 20261023-200308-twist-image-release-gate-854444 fails in LLVM compilation under the 8 GiB address limit; no test executed.

Replacement 20261023-211022-twist-image-gate-v2-5c8a60 builds separately, then runs serial tests under the original 8 GiB / 1,200 s limit. All six compile/test exits are zero: 4,488 library tests passed, zero failed, 119 ignored; examples passed 1 and 3 tests; all 57 native tests passed. Library execution took 977.90 seconds; the job completed at 21:54 UTC on 2026-10-23. Saved receipts and logs establish these outcomes after migration; tests were not repeated. Compilation is not performance evidence and changes no scientific arm's frozen source or resource envelope.

PR head c7cecbd07f437b600b5893be61ad0eedf38d2368 fails Clippy's manual_is_multiple_of in untouched curve_traits/walk.rs and prime_small_x.rs; QuarticTwistSymmetry Lean CI fails toolchain extraction. Logs are retained; neither check is bypassed or credited as passing. The merge conflict was referred to Copilot. Every later head requires its own review and CI.

Missing claim gates

No complete GAE operation price, 0/0.1/1 GAE band, protocol-matched strongest rho reference, independent sampling of target scalars, or growing-size inference follows from these fixture timings. Magma, FGb, Singular and OpenF4 were not reproduced. msolve is generally faster on the GB component; it has not been integrated and priced as an external complete DLP solver.

Accordingly this directory has no asserted VALIDATION.json and contributes no new row to the gated leaderboard or benchmark index. Honest finite positive and negative observations remain in the report with their exact boundary. See the unresolved problem (../../../problems/quartic-twist-polynomial-ic-pricing.md) and the raw evidence (ARTIFACTS.md).